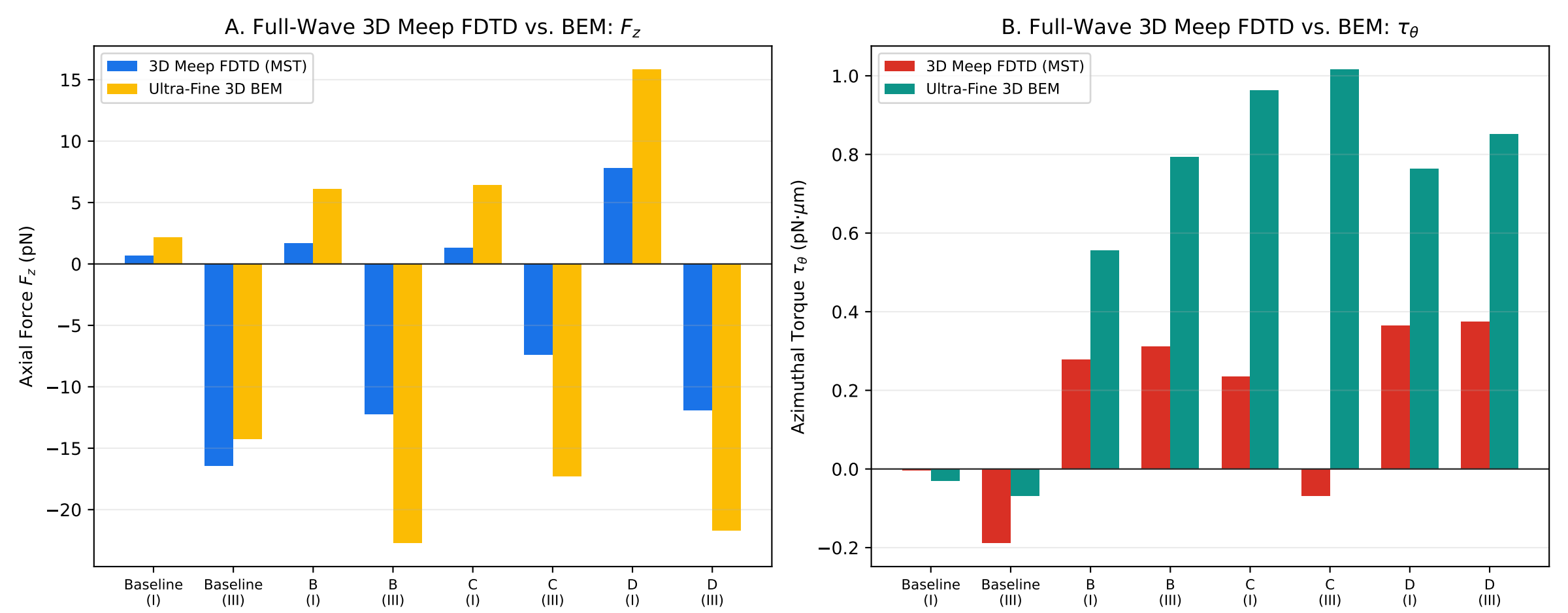
A. Full-Wave 3D Meep FDTD vs. BEM: F_z
3D Meep FDTD (MST)
Ultra-Fine 3D BEM
Axial Force F_z (pN)
15
10
5
0
−5
−10
−15
−20
Baseline (I)
Baseline (III)
B (I)
B (III)
C (I)
C (III)
D (I)
D (III)
B. Full-Wave 3D Meep FDTD vs. BEM: τ_θ
3D Meep FDTD (MST)
Ultra-Fine 3D BEM
Azimuthal Torque τ_θ (pN·μm)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Baseline (I)
Baseline (III)
B (I)
B (III)
C (I)
C (III)
D (I)
D (III)